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**Spatial inequality in the accessibility of public libraries in
New York City**

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GGIS 224
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10 May 2025

Section 1. Background and introduction

Public libraries play a vital role in providing equal access to information, digital resources, and community services. In New York City (NYC), the three public library systems: New York Public Library, Brooklyn Public Library, and Queens Public Library—collectively serve nearly 40 million residents annually, highlighting their vital role in daily life (NYC Mayor's Office, 2024). However, spatial disparities in access remain a persistent concern, particularly in communities with lower incomes or higher proportions of racial minorities. In particular, children in underprivileged neighborhoods are less likely to read at grade level because they have less access to books and learning environments, which can worsen already-existing social inequities (Dusenbury et al., 2006).

Previous studies have found disparities in library access and use according to neighborhood attributes like income, race, and digital infrastructure (Dusenbury et al., 2006; Pi Tech, 2023). The main research question is: ***Are library services in New York City equally accessible across neighborhoods with different income levels and racial compositions?***

According to my first hypothesis, neighborhoods with lower median household incomes and higher proportions of Black or Hispanic residents will have lower access to public libraries. This assumption is supported by literature showing that library accessibility and usage are shaped by broader structural inequalities—particularly the digital divide, literacy gaps, and inadequate infrastructure in communities (Jaeger et al., 2012).

Understanding these disparities is critical for informing equitable urban planning and addressing long-standing structural barriers. This study applies geospatial and statistical methods to examine the spatial distribution of library accessibility and offers insights for improving equity in public infrastructure planning.

Section 2. Dataset

Four primary sources of data were used for this project: (1) a shapefile of census tract boundaries in New York City; (2) a CSV/GeoJSON file containing the locations of all public library branches across the five boroughs; (3) tract-level demographic and socioeconomic indicators from the American Community Survey (ACS) 5-Year Estimates; and (4) OpenStreetMap (OSM) road network data used to compute walking-based travel times and isochrone catchments. (More information in appendix).

Section 3. Methods

I conducted a multi-step workflow involving data wrangling, spatial operations, network-based modeling, and statistical analysis. All tasks were performed using R.

1 Data Wrangling – I compiled a spatial data frame (**NYC_data**) by merging: **Census tracts** from `tidycensus::get_acs()`, projected to EPSG:2263. **Demographics** (Black/Hispanic percentages, population, income) from ACS, joined via **GEOID**, with group shares computed as proportions of total population. **Library locations** from NYC Open Data, filtered and converted to **sf** points.

2 Accessibility Modeling Using r5r – Using **r5r**, I computed walking times from tract centroids to the nearest library: Tract centroids were derived with `st_centroid()`. A routable transport network was built from a local OSM **.pbf** file. `r5r::travel_time_matrix()` calculated median walking times **traveltimemp50**.

I also attempted to use 30-minute isochrones but failed due to computational limits thus were excluded from final modeling.

3 Spatial Visualization – I used **tmap** to produce: A choropleth map of **travel_time_p50** by quantile. Choropleth maps overlaying race/income. A kernel density map of library locations using point density estimation.

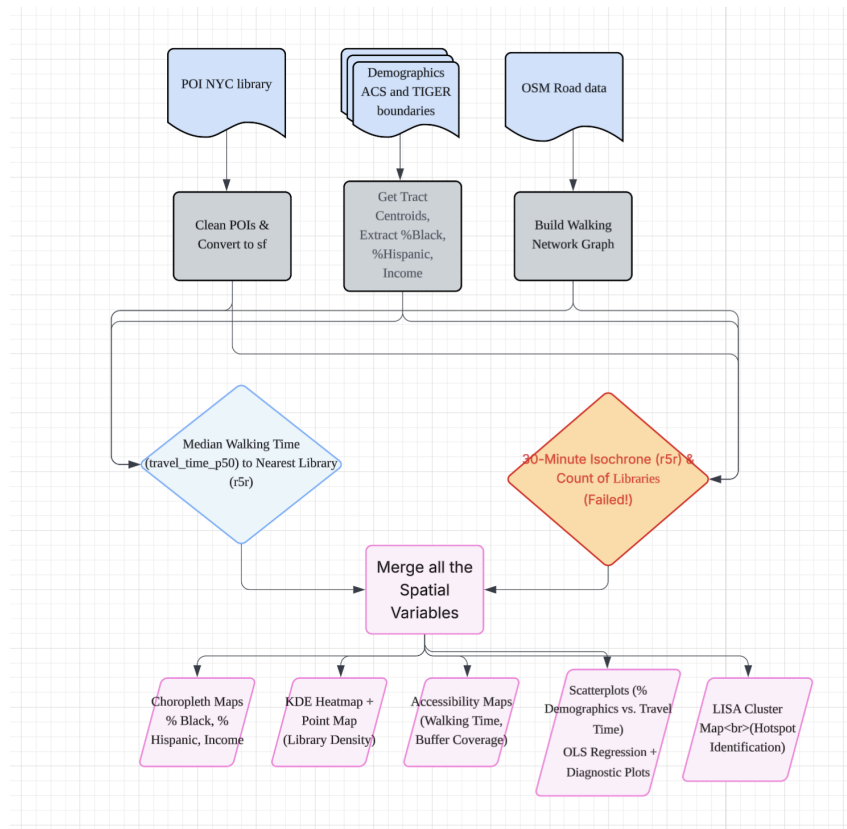
4 Spatial Autocorrelation (LISA) – Using **spdep**, I performed Local Moran's I analysis: A Queen contiguity matrix defined neighborhood structure. LISA clusters (e.g., High-High, Low-Low) were identified and mapped, highlighting spatial clustering in access times.

5 Linear Regression Modeling – I ran an OLS regression to assess how accessibility relates to race and income:

$$\text{travel_time_p50} = \beta_0 + \beta_1 \cdot \text{pct_black} + \beta_2 \cdot \text{pct_hispanic} + \beta_3 \cdot \text{income} + \varepsilon$$

Model assumptions (normality, linearity, homoscedasticity) were checked via residual plots and Q-Q plots. Scatterplots with `geom_smooth()` visualized key relationships.

6 Conceptual Diagram



Section 4. Results

4.1 Kernel Density of Public Libraries in NYC

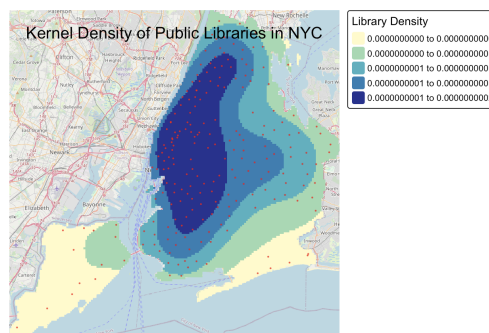


Figure 1. Kernel Density of Public Libraries in NYC

Kernel density estimation (KDE) reveals clusters of public library locations across New York City. This figure provides an initial spatial overview of public library availability across the city.

- **Hotspots** of library density are observed in **lower Manhattan** and **central Brooklyn**, areas that historically serve larger populations and benefit from concentrated civic infrastructure.
- **Low-density regions**, such as **southern Staten Island** and **eastern Queens**, are visibly underserved based on library location alone.

4.2 Tract-Level Library Accessibility and visualizing the social data

NYC Tract Accessibility to Nearest Library (5-min Intervals)

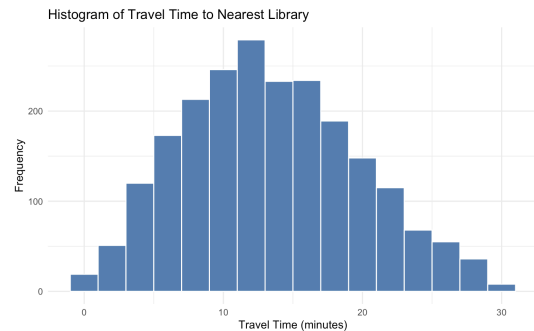
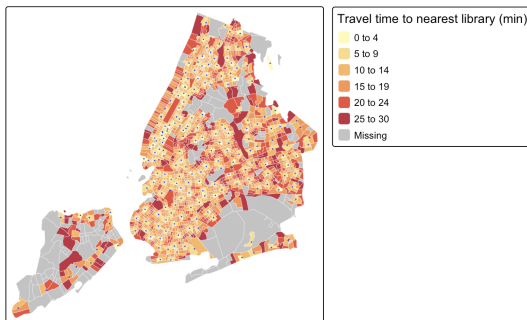


Figure 2. NYC Tract Accessibility to Nearest Library **Figure 3.** Histogram of Travel Time

To understand the spatial distribution of library accessibility across New York City, I mapped the median walking time from each census tract centroid to the nearest public library. Figure 2 shows the spatial variation in travel time, grouped into 5-minute intervals. The histogram in Figure 3 illustrates the overall distribution, with most tracts experiencing 10–15 minutes of walking time.

Notably, neighborhoods in southern Staten Island, southeastern Queens, and parts of the Bronx show longer access times (20–30 minutes), while central Manhattan and Brooklyn have the shortest travel times due to higher library density and walkability. These patterns reflect disparities in urban infrastructure and service.

NYC Census Tract % Black Population

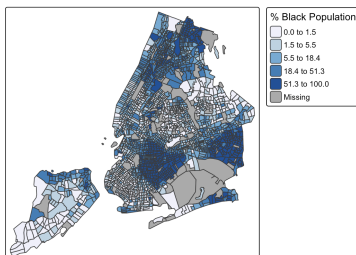


Figure 4

NYC Census Tract % Hispanic Population

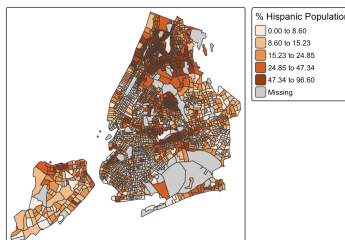


Figure 5

NYC Census Tract Median Household Income

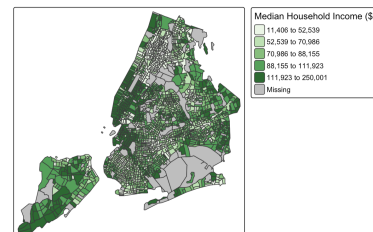


Figure 6

In addition to accessibility measures, I also visualized key sociodemographic indicators to understand the spatial patterns of race and income across New York City.

Figure 4 shows the distribution of the percentage of Black residents, with higher concentrations in central Brooklyn, southeastern Queens, and parts of the Bronx.

Figure 5 maps the percentage of Hispanic or Latino residents, which are more prominent in northern Manhattan, the South Bronx, and western Queens.

Figure 6 displays median household income, highlighting wealthier tracts in Manhattan's core, parts of Queens, and northwestern Brooklyn, while lower-income neighborhoods are concentrated in the South Bronx, eastern Brooklyn, and central Queens.

4.3 Spatial Autocorrelation of Library Travel Time

To assess spatial clustering in library accessibility, I conducted global and local spatial autocorrelation analysis.

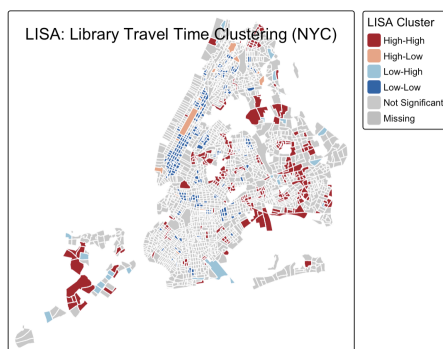


Figure 7

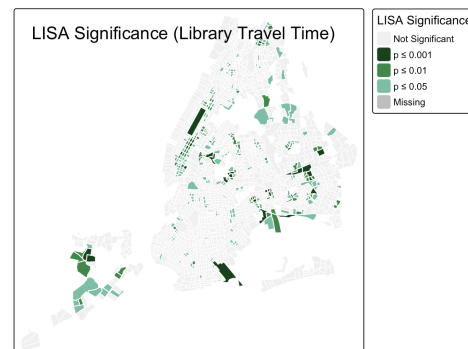


Figure 8

Figure 7. LISA cluster map of travel time in NYC. Red areas indicate tracts with high travel time surrounded by similar underserved areas (High-High clusters), while blue areas indicate Low-Low clusters. High-High are mainly concentrated in **southern Queens, eastern Brooklyn, and Staten Island**. These regions are underserved and may lack nearby library facilities or walkable routes. where tracts with short library access times are mostly located in **Manhattan and central Brooklyn**, areas historically better served by public infrastructure.

Figure 8. LISA significance map shows the statistical significance of these clusters, with many tracts significant at $p \leq 0.01$ or $p \leq 0.001$.

```
4 > moranI
5 global_moran <- moran(map_data_clean$travel_time_p50, lw_queen,
6                       length(nb_queen), Szero(lw_queen))
7 global_moran$I
8
9 [1]
10
[1] 0.3824567
```

The Global Moran's I value is 0.3825, indicating a moderate positive spatial autocorrelation.

These results support the hypothesis that library access is not spatially equal across New York City. Significant clusters of poor access coincide with lower-income and minority-heavy neighborhoods like southern Queens and parts of the Bronx, highlighting spatial inequities in cultural infrastructure.

4.4 Regression Analysis: Modeling Access Inequality

To quantify the relationship between demographic, socioeconomic characteristics and library access, an Ordinary Least Squares (OLS) regression is used.

Only the % of the Hispanic population showed a statistically significant ($p < 0.05$) negative association with travel time that higher proportions were associated with slightly shorter library access times, which contrasts with the initial hypothesis(Fig12). Neither the percentage of Black residents nor income showed significant effects, and the overall R-squared value of the model was very low (0.0036), indicating limited explanatory power.

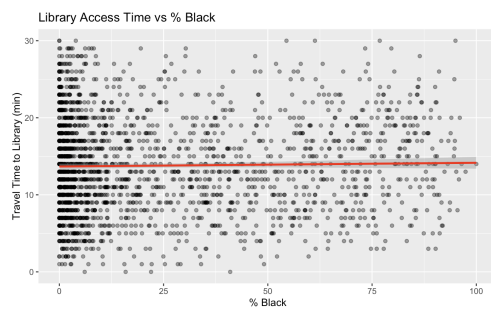


Figure 9a

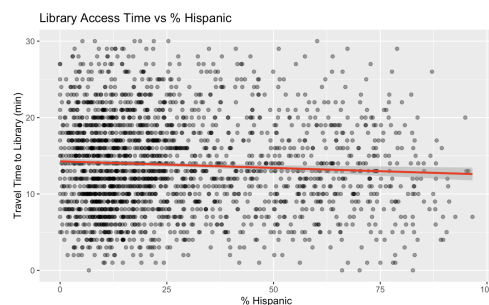


Figure 9b

Figure 9a: A scatterplot of % Black vs. `travel_time_p50` with a fitted regression line showed a flat trend, confirming the non-significant relationship.

Figure 9b: A scatterplot of % Hispanic vs. `travel_time_p50` showed a mild downward slope, aligning with the negative coefficient found in the model.

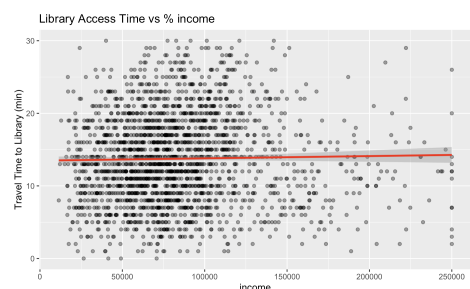


Figure 9c

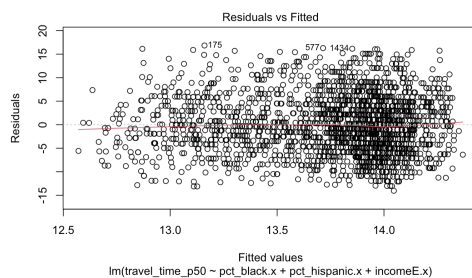


Figure 10

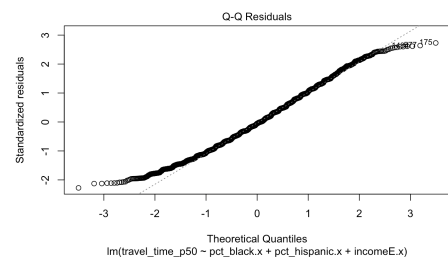


Figure 11

```
Call:
lm(formula = travel_time_p50 ~ pct_black_z + pct_hispanic_z +
    incomeE_z, data = lisa_data_std)

Residuals:
    Min       1Q   Median       3Q      Max
-14.046  -4.620  -0.335   4.188  16.840

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  13.74418    0.13437  102.286   <2e-16 ***
pct_black_z    0.06251    0.14493   0.431   0.6663
pct_hispanic_z -0.36618    0.15646  -2.340   0.0194 *
incomeE_z     -0.01404    0.16072  -0.087   0.9304
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.176 on 2109 degrees of freedom
(74 observations deleted due to missingness)
Multiple R-squared:  0.003582, Adjusted R-squared:  0.002165
F-statistic: 2.527 on 3 and 2109 DF, p-value: 0.05579
```

Figure 12

Figure 9c: A scatterplot of **Income** vs. **travel_time_p50** revealed a weak and slightly upward trend, consistent with its non-significant and near-zero regression coefficient.

Figure 10: The residuals vs. fitted values plot did not show major violations of homoscedasticity, though the spread suggested high variability.

Figure 11: The Q-Q plot of residuals demonstrated approximate normality, with slight deviations in the tails.

These statistical results suggest that there is no strong or consistent linear relationship between race/income variables and library access time.

However the LISA analysis presented earlier did reveal spatial clustering patterns, suggesting that inequality may manifest more locally or non-linearly.

4.5 limitations of analysis.

Limited Explanatory Power of Statistical Model

The OLS regression model showed very low explanatory power ($R^2 = 0.0036$), suggesting that the selected variable can explain only a very small portion of the variance. Additionally, only the % Hispanic variable showed statistical significance, and in a direction contrary to the initial hypothesis. This highlights potential model misspecification or the presence of unmeasured factors (e.g., land use, transportation barriers, or institutional policies) that influence accessibility but were not included in the analysis. When selecting social variables and using statistical models, I should be more comprehensive and considerate.

Isochrone Method Limitations

This study only used a walking-based network to estimate travel time, it did not include full isochrone-based service area modeling to count library facilities within a walkable region. This decision was made due to the high computational cost of generating isochrones, but it has reduced the completeness of accessibility modeling as number within a certain minute isochrone is also an important factor in accessing accessibility.

No Control for Population Density or Demand

The analysis modeled access to libraries but did not control for population density, service demand, or library size/capacity. Some areas with fewer libraries may still be adequately served if the population is small, and vice versa.

Section 5. Conclusions

This project studied whether access to public libraries in NYC is distributed equitably across census tracts of different income and racial compositions. The spatial analysis showed regional disparities—particularly in areas like southern Queens and the Bronx, but the statistical model showed limited explanatory power.

The LISA cluster map and Moran's I (0.3825) indicated significant spatial autocorrelation in travel time to the nearest library. However, the multiple linear regression model found only a weak negative relationship between accessibility and the percentage of Hispanic residents. No significant relationships were found for income or Black population percentages.

The initial hypothesis—expecting poorer and majority-minority neighborhoods to have lower access—was only partially supported.

There are several limitations to this analysis. Isochrone-based service counts were omitted, possibly underrepresenting accessibility in dense areas. The possible ACS estimation errors, and the simplicity of the regression model also limit validity.

Future research will include more detailed modeling, non-linear or spatial regression approaches, and incorporate additional predictors such as transit coverage, age structure. Despite its limitations, this project underscores persistent spatial inequalities in public infrastructure and offers a guide for more targeted planning and policy making.

Bibliography

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Appendix – Data used and detailed information

New York City Library Branches

Official Name: Library Branches – NYC Public Libraries

Source: NYC Open Data <https://data.cityofnewyork.us/Business/LIBRARY/feuq-due4>

Geographic Coverage: Five boroughs of New York City

Description: Includes names, addresses, and coordinates of all public library branches

NYC Census Tract Boundaries

Official Name: 2020 TIGER/Line® Shapefiles – Census Tracts

Source: US Census Bureau

<https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html>

Geographic Scale: Census Tracts (within NYC)

Demographic and Socioeconomic Data

Official Name: American Community Survey (ACS) 5-Year Estimates (2018–2022)

Source: United States Census Bureau, via **tidycensus** R package

URL: <https://www.census.gov/programs-surveys/acs>

Road Network and Transit Data

Official Name: OpenStreetMap PBF – New York City

Source: OpenStreetMap contributors, processed via **r5r**

URL: <https://download.geofabrik.de/north-america/us/new-york.html>

Timescale: Latest available extract (April 2025)

Use: Used to compute walking travel time matrices and isochrones